GABRIELLA ESEREOMO AVBAYERU

GIS & Remote Sensing · Geospatial Analysis

Ibadan, Nigeria · +234 805 512 7802 · gabriellaavbayeru@gmail.com

Portfolio: mygabbbie.github.io/GIS-Portfolio · GitHub: github.com/MyGabbbie

SUMMARY

Master's student in Geoinformation Science with hands-on, project-based experience in land cover classification, change detection, hydrological modelling, and spectral index analysis using ArcGIS Pro, QGIS, and Google Earth Engine. Seeking GIS, geospatial analysis, or remote sensing research roles.

EDUCATION

MSc, Geoinformation Science — University of Ibadan *Expected 2027*

BSc, Urban and Regional Planning — Federal University of Technology, Akure *2024*

SKILLS

GIS Software: ArcGIS Pro, ArcMap, QGIS, Google Earth Engine

Remote Sensing & Analysis: Land cover classification, change detection, Markov transition modelling, landscape metrics, spectral indices (NDVI, SAVI, NDWI, NDBI), hydrological & watershed delineation, raster reclassification, spatial overlay & geoprocessing

Data & Programming: Python (geospatial libraries), Excel / tabular analysis, vector & raster data management

Communication: Technical & research writing, independent research design, cartographic production, data storytelling

Languages: English (Fluent)

PROJECTS

Spatiotemporal LULC Dynamics & Urban Growth Drivers, Edo State, Nigeria *1999–2024*
Coursework assignment — Google Earth Engine, ArcGIS Pro

- Classified six epochs of Landsat imagery into five land cover classes; built a Markov transition matrix and six landscape structural metrics to explain conversion patterns and fragmentation.
- Found built-up area grew 37.04% against a 96.28% population increase — evidence of urban densification rather than sprawl — and mapped five distinct urban growth patterns across the state's major cities.

Eleyele Reservoir — Vegetation Encroachment & Water Surface Dynamics *2017–2025*
Coursework assignment — ArcGIS (ArcMap)

- Analyzed nine years of vector, raster, and monthly time-series data to track aquatic vegetation encroachment and water surface change.
- Quantified a 59% decline in average open-water surface area (41.4 ha → 17.1 ha) and produced vegetation persistence and seasonal dynamics maps.

Ibadan North — Multi-Index Land Surface Mapping *2017*
Independent project — Google Earth Engine

- Computed and cross-compared NDVI, SAVI, NDWI, and NDBI to characterize vegetation, soil, moisture, and built-up surface patterns for the same study area and date.

Ibarapa East — Drainage Basin Delineation
Independent project — ArcGIS Pro, SRTM DEM (USGS)

- Delineated five sub-basins and their stream network from raw SRTM elevation data using ArcGIS Pro's hydrology toolset (Fill, Flow Direction, Flow Accumulation, Watershed).

LEADERSHIP & VOLUNTEER EXPERIENCE

Welfare Director — Enactus FUTA *2024*

Vice President — Department of Urban and Regional Planning Students' Association, FUTA *2022–2023*